

CE205: Finite Element Method

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25053

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1 Problem Description

Analysis of the effect of a circular through-hole in a rectangular cantilevered beam. The length of the beam is $L = 127$ mm, and its height H is 25.4 mm, giving a slenderness ratio of 5. The hole is of radius $r = 0.5H = 12.70$ mm, and its center is located along the mid-line of the beam at a distance of $x_c = 0.5L = 63.5$ mm from the cantilevered end, as shown in Fig. 1. The beam is subjected to a single downward vertical load $P = 1000$ N at its top-right corner. The out-of-plane beam thickness t is 5 mm. For aluminum, $E = 70$ GPa and $\nu = 0.33$.

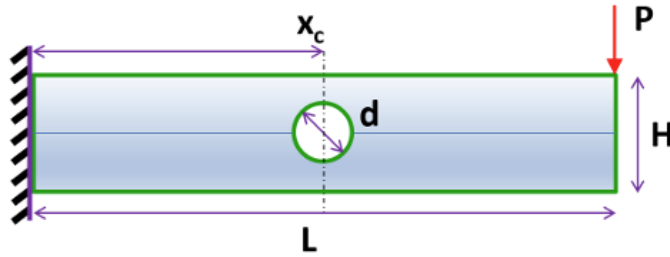


Figure 1: A cantilever beam with a circular hole

2 Implementation

2.1 isoparamQ4stiffness.m

To test that the Q4 isoparametric element works, it is used (with full integration) in place of the rectangular Q4 elements in HW 4 with mesh parameters $n_h = 60$ and $n_v = 2$, and the displacement values are compared. It is found that the displacements are identical to within numerical precision.

	Displacement (m)
Rectangular Q4	-6.2872501281940939e-4
Isoparametric Q4	-6.2872501280459051e-4
Difference	1.4819e-14

Table 1: Comparison of displacement for rect Q4 and isoparam Q4

For a typical rectangular element with nodes $(0, 0)$, $(2, 0)$, $(2, 1)$ and $(0, 1)$, the value of the Jacobian J is found to be 0.5 at all integration points. This is equal to $\frac{1}{4}$ times the area of the rectangular element, as expected. The evaluation is done using **jacobianCheckQ4.m**.

2.2 isoparamT6stiffness.m

To test that the T6 isoparametric element works, it is used in place of the rectangular Q4 elements in HW 4, and a mesh convergence study is done. The plateauing of the curve in Fig. 2 shows convergence and verifies the element's working.

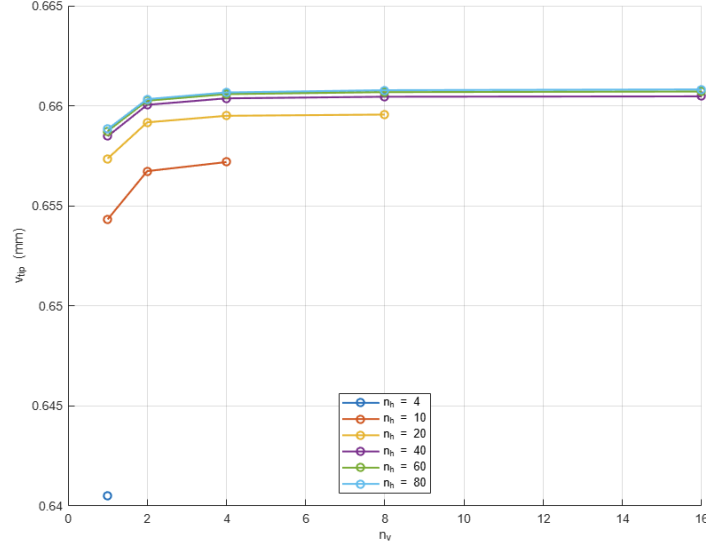


Figure 2: Mesh convergence study for HW 4 problem with T6 element

For a straight-sided triangular element with side-nodes at mid-points of edges i.e. a triangle with nodes $(0, 0)$, $(1, 0)$, $(\frac{1}{2}, \frac{\sqrt{3}}{2})$, $(\frac{1}{2}, 0)$, $(\frac{3}{4}, \frac{\sqrt{3}}{4})$, $(\frac{1}{4}, \frac{\sqrt{3}}{4})$, the value of Jacobian J is found to be 0.866, which is twice the area of triangle, as expected. The evaluation is done using **jacobianCheckT6.m**.

2.3 Mesh Generation

The mesh was generated using *Gmsh*. The detailed pictures of the meshes for both Q4 and T6 are given below. Different geometries are used for either element. The mesh file is extracted in *.msh* extension. Relevant information, such as coordinates of a node, nodes of an element, and nodes along the boundary of the beam, is derived from this using **extractQ4.m** and **extractT6.m** respectively.

- Q4

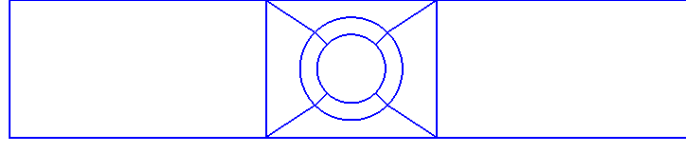


Figure 3: Geometry for Q4

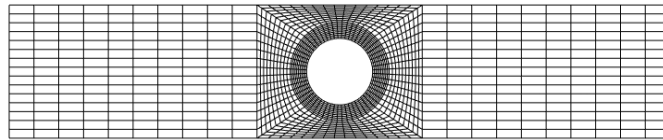


Figure 4: Q4 mesh (converged)

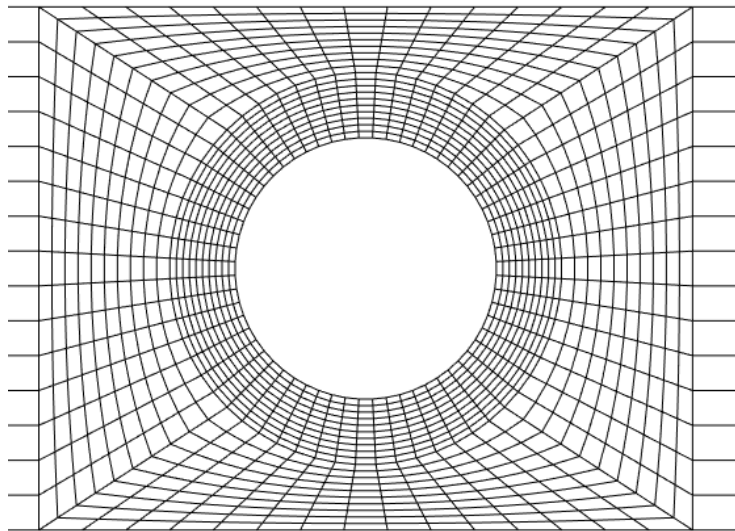


Figure 5: Zoomed around hole in Q4 mesh (converged)

- T6

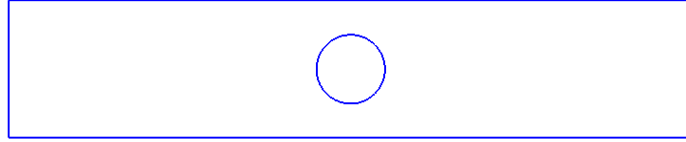


Figure 6: Geometry for T6

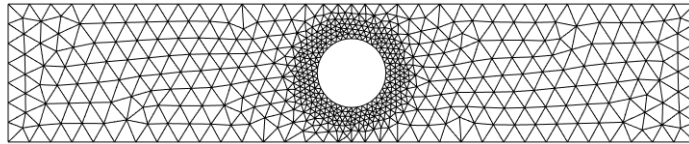


Figure 7: T6 mesh (converged)

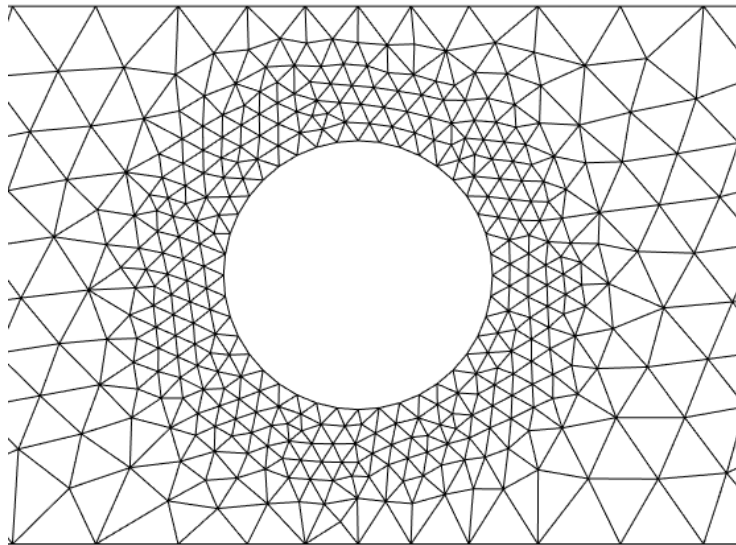


Figure 8: Zoomed around hole in T6 mesh (converged)

2.4 Stress Computation

As in HW4, the point load is equally divided between the top-right and bottom-right nodes.

2.4.1 Isoparametric Q4

At Gauss Points: For each element, stresses at Gauss Points are computed using `elemStressGPQ4.m`.

At Nodes: To obtain stresses at nodes, the strategy is to extrapolate the stress values from the Gauss points to the nodes, then average the values obtained from all the elements sharing that node to obtain the nodal stresses.

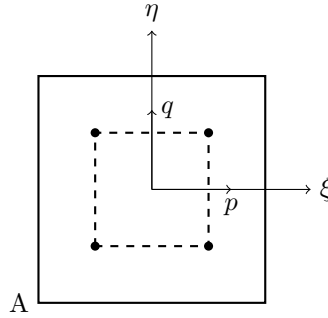


Figure 9: Extrapolation of stress in Q4

For the Q4 element, the 2×2 Gauss points are located at $\xi = \eta = \pm \frac{1}{\sqrt{3}}$. Let p and q be scaled version of the $\xi - \eta$ system such that they have the same origin but with $p = q = 1$ when $\xi = \eta = \frac{1}{\sqrt{3}}$. In other words, let $p = \sqrt{3}\xi$ and $q = \sqrt{3}\eta$. The extrapolated stress components are given by (σ is a typical stress component)

$$\sigma = \sum_{i=1}^4 \bar{N}_i \sigma_i,$$

where σ_i are the values of the stress components at the Gauss points, and

$$\bar{N}_i = \frac{1}{4}(1 \pm p)(1 \pm q)$$

Thus, the stress components at node A in Fig. 9 are obtained by substituting $p = q = -\sqrt{3}$ in the above formula. Similarly, the contribution from neighbouring elements sharing node A is summed and divided by

the number of elements sharing node A. The computation is done using **elemStressNodalQ4.m**.

$$\sigma_{nodeA} = \frac{\sum_{i=1}^n \sigma_{A,i}}{n}$$

where n is the number of elements sharing node A.

Hoop stress and von Mises stress: For the nodes on the edge of the hole, hoop stress is calculated using stress transformation, given by the formula:

$$\sigma_{\theta\theta} = \sigma_{xx}\sin^2(\theta) + \sigma_{yy}\cos^2(\theta) - \sigma_{xy}\sin(2\theta)$$

For a plane stress condition,

$$\sigma_{zz} = \sigma_{yz} = \sigma_{xz} = 0$$

Hence, the von Mises stress is calculated as:

$$\sigma_{vm} = \sqrt{\sigma_{xx}^2 - \sigma_{xx}\sigma_{yy} + \sigma_{yy}^2 + 3\sigma_{xy}^2}$$

At (x,y): For the calculation of stress concentration factor K , σ_{xx}^o is to be evaluated at (x, y) in the hole-free beam with converged mesh. This (x, y) is the same location along the edge of the hole, where hoop stress is maximum.

To find the mapped coordinates (ξ, η) from the physical coordinate (x, y) , we solve a system of nonlinear equations in ξ and η . The equations are solved numerically using the *Newton-Raphson* method. The steps involved are:

Step 1: Start with the initial condition $(\xi_0, \eta_0) = (0, 0)$ i.e. the center of the mapped element.

Step 2: Compute the shape functions and their derivatives at the current guess.

Step 3: Use the shape functions to compute the approximate location (x_k, y_k) corresponding to the current mapped coordinates:

$$x_k = \sum_{i=1}^4 N_i(\xi, \eta)x_i \quad y_k = \sum_{i=1}^4 N_i(\xi, \eta)y_i$$

Here, x_i and y_i are the nodal coordinates of the element.

Step 4: Calculate the difference between the target (x, y) and approximated (x_k, y_k) :

$$F = \begin{bmatrix} x - x_k \\ y - y_k \end{bmatrix}$$

Step 5: Compute Jacobian matrix, solve for the correction, and update the guess.

$$\Delta = \begin{bmatrix} \Delta\xi \\ \Delta\eta \end{bmatrix} = J^{-1}F$$

$$\xi_{k+1} = \xi_k + \Delta\xi \quad \eta_{k+1} = \eta_k + \Delta\eta$$

Step 6: If $\|\Delta\|$ is less than the specified tolerance and ξ_{k+1} and η_{k+1} is between -1 and 1 , the process is stopped. Otherwise, repeat from step 2 with the updated guess.

The above computation is done using **findElementQ4.m**, which gives the corresponding (ξ, η) coordinate of (x, y) and the element in which the physical coordinate lies.

The stresses at this mapped coordinate are evaluated by extrapolating the stresses at the nodes to the mapped coordinate using the displacement shape functions.

$$\sigma(\xi, \eta) = \sum_{i=1}^4 N_i \sigma_{node,i}$$

where $\sigma_{node,i}$ is the stress at i^{th} node of the element. This computation is done using **stressInElementQ4.m**.

2.4.2 Isoparametric T6

At Gauss Points: For each element, stresses at Gauss Points are computed using **elemStressGPT6.m**.

At Nodes: To obtain stresses at nodes, the strategy is to extrapolate the stress values from the Gauss points to the nodes, then average the values obtained from all the elements sharing that node to obtain the nodal stresses.

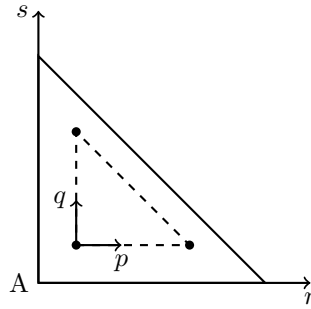


Figure 10: Stress extrapolation in T6

For the T6 element, the interior Gauss points are located at $(r, s) = (\frac{1}{6}, \frac{1}{6}), (\frac{2}{3}, \frac{1}{6})$ and $(\frac{1}{6}, \frac{2}{3})$. Let p and q be scaled version of the $r - s$ system

such that they have the origin at $(r, s) = (\frac{1}{6}, \frac{1}{6})$. Then $(p, q) = (1, 0)$ at $(r, s) = (\frac{2}{3}, \frac{1}{6})$ and $(p, q) = (0, 1)$ at $(r, s) = (\frac{1}{6}, \frac{2}{3})$. In other words, let $p = 2r - \frac{1}{3}$ and $q = 2s - \frac{1}{3}$. The extrapolated stress components are given by (σ is a typical stress component)

$$\sigma = \sum_{i=1}^3 \bar{N}_i \sigma_i,$$

where σ_i are the values of the stress components at the Gauss points, and

$$\bar{N}_1 = 1 - p - q \quad \bar{N}_2 = p \quad \bar{N}_3 = q$$

Thus, the stress components at node A in Fig. 10 are obtained by substituting $p = q = -\frac{1}{3}$ in the above formula. Similarly, the contribution from neighbouring elements sharing node A is summed and divided by the number of elements sharing node A. The computation is done using **elemStressNodalT6.m**.

$$\sigma_{nodeA} = \frac{\sum_{i=1}^n \sigma_{A,i}}{n}$$

where n is the number of elements sharing node A.

Hoop stress and von Mises stress: For the nodes on the edge of the hole, hoop stress is calculated using stress transformation, given by the formula:

$$\sigma_{\theta\theta} = \sigma_{xx} \sin^2(\theta) + \sigma_{yy} \cos^2(\theta) - \sigma_{xy} \sin(2\theta)$$

For a plane stress condition,

$$\sigma_{zz} = \sigma_{yz} = \sigma_{xz} = 0$$

Hence, the von Mises stress is calculated as:

$$\sigma_{vm} = \sqrt{\sigma_{xx}^2 - \sigma_{xx}\sigma_{yy} + \sigma_{yy}^2 + 3\sigma_{xy}^2}$$

At (x,y): For the calculation of stress concentration factor K , σ_{xx}^o is to be evaluated at (x, y) in the hole-free beam with converged mesh. This (x, y) is the same location along the edge of the hole, where hoop stress is maximum.

To find the mapped coordinates (r, s) from the physical coordinate (x, y) , we solve a system of nonlinear equations in r and s . The equations are solved numerically using the *Newton-Raphson* method. The steps involved are:

Step 1: Start with the initial condition $(r_0, s_0) = (\frac{1}{3}, \frac{1}{3})$ i.e. the centroid of the mapped element.

Step 2: Compute the shape functions and their derivatives at the current guess.

Step 3: Use the shape functions to compute the approximate location (x_k, y_k) corresponding to the current mapped coordinates:

$$x_k = \sum_{i=1}^6 N_i(r, s) x_i \quad y_k = \sum_{i=1}^6 N_i(r, s) y_i$$

Here, x_i and y_i are the nodal coordinates of the element.

Step 4: Calculate the difference between the target (x, y) and approximated (x_k, y_k) :

$$F = \begin{bmatrix} x - x_k \\ y - y_k \end{bmatrix}$$

Step 5: Compute Jacobian matrix, solve for the correction, and update the guess:

$$\Delta = \begin{bmatrix} \Delta r \\ \Delta s \end{bmatrix} = J^{-1} F$$

$$r_{k+1} = r_k + \Delta r \quad s_{k+1} = s_k + \Delta s$$

Step 6: If $\|\Delta\|$ is less than the specified tolerance and r_{k+1} and s_{k+1} is between 0 and 1 and $r + s \leq 1$, the process is stopped. Otherwise, repeat from step 2 with the updated guess.

The above computation is done using **findElementT6.m**, which gives the corresponding (r, s) coordinate of (x, y) and the element in which the physical coordinate lies.

The stresses at this mapped coordinate are evaluated by extrapolating the stresses at the nodes to the mapped coordinate using the displacement shape functions.

$$\sigma(r, s) = \sum_{i=1}^6 N_i \sigma_{node, i}$$

where $\sigma_{node, i}$ is the stress at i^{th} node of the element. This computation is done using **stressInElementT6.m**.

3 Results

3.1 Isoparametric Q4

3.1.1 Convergence study of beam with hole

N_h is the number of elements along the edge of the hole. N_{el} is the total number of elements. v^* is the tip selection of the bottom-right corner of the beam. $\sigma_{\theta\theta}^{max}$ is the maximum hoop stress along the edge of the hole. Highlighted row denotes

converged mesh. v^* and $\sigma_{\theta\theta}^{max}$ are plotted against N_h and N_{el} to find converged mesh (with full integration).

N_h	N_{el}	v^* (mm)	$\sigma_{\theta\theta}^{max}$ (MPa)
8	220	1.4173	61.1338
12	330	1.4556	101.1057
16	440	1.4717	108.9226
20	550	1.4807	122.8548
40	1100	1.4944	141.8797
60	1650	1.4974	148.0136
80	2200	1.4985	150.0448
100	2750	1.4999	150.8253

Table 2: Mesh convergence study for beam with hole (Q4)

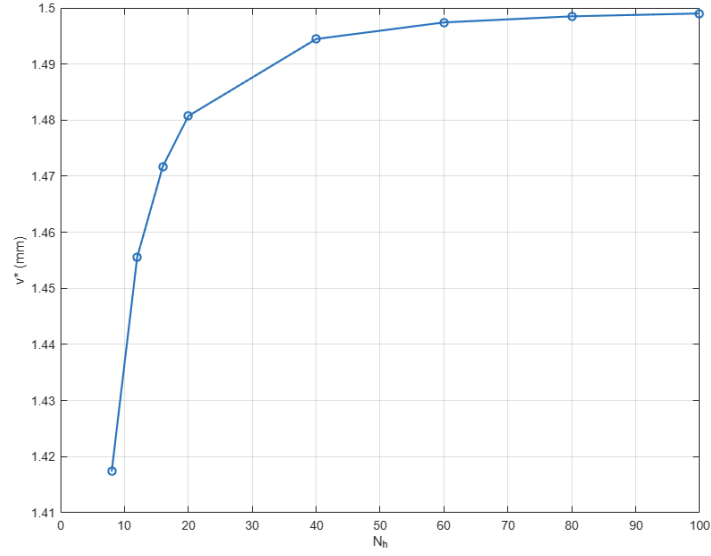


Figure 11: v^* vs N_h (Q4)

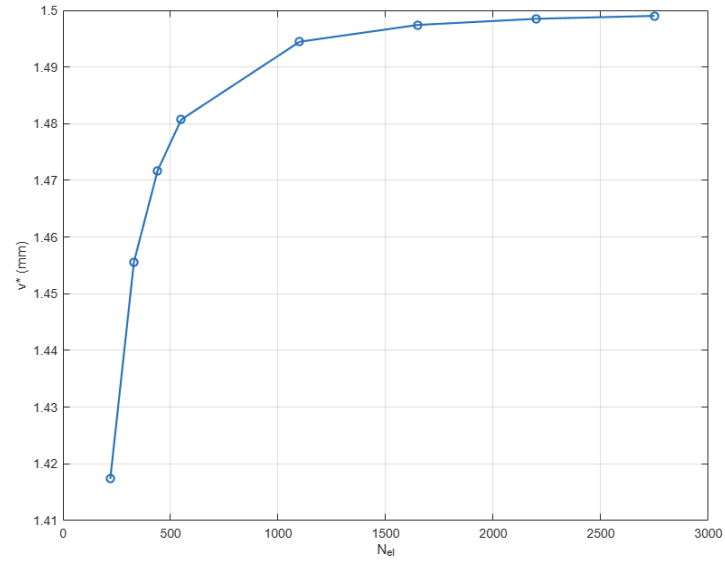


Figure 12: v^* vs N_{el} (Q4)

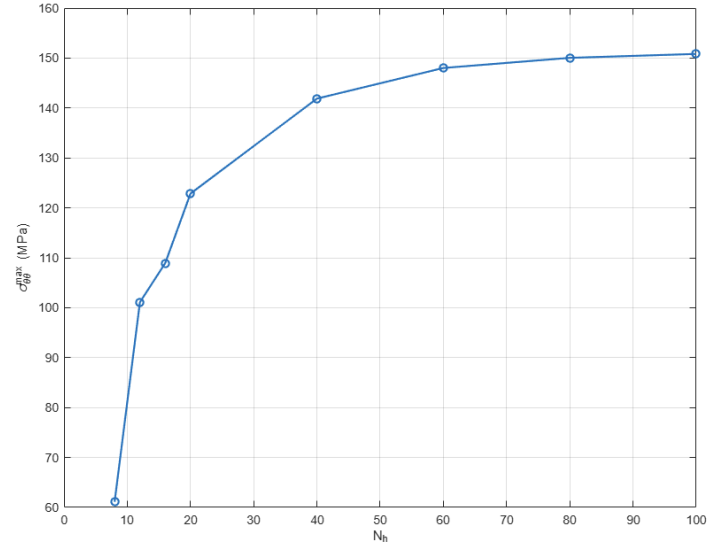


Figure 13: $\sigma_{\theta\theta}^{max}$ vs N_h (Q4)

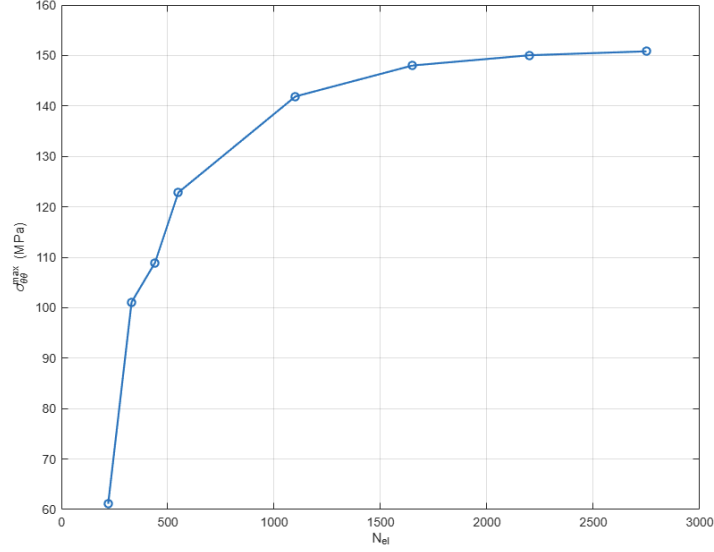


Figure 14: $\sigma_{\theta\theta}^{max}$ vs N_{el} (Q4)

3.1.2 Convergence study of hole-free beam

n_h is the number of elements along the horizontal edge. n_v is the number of elements along the vertical edge. v^o is the deflection of the bottom-right corner of the beam. The total number of elements is $n_h \times n_v$. Highlighted row denotes the converged mesh. v^o is plotted against n_v for every n_h to find converged mesh (with full integration).

n_h	n_v	v^o (mm)
4	1	0.8605
10	1	1.2096
	2	1.2967
	4	1.3253
20	1	1.2841
	2	1.3868
	4	1.4211
	8	1.4307
40	1	1.3041
	2	1.4118
	4	1.4485
	8	1.4593
	16	1.4624
60	1	1.3079
	2	1.4166
	4	1.4539
80	8	1.4653
	16	1.4687
	1	1.3092
	2	1.4183
	4	1.4558
	8	1.4675
	16	1.4713

Table 3: Mesh convergence study for beam without hole (Q4)

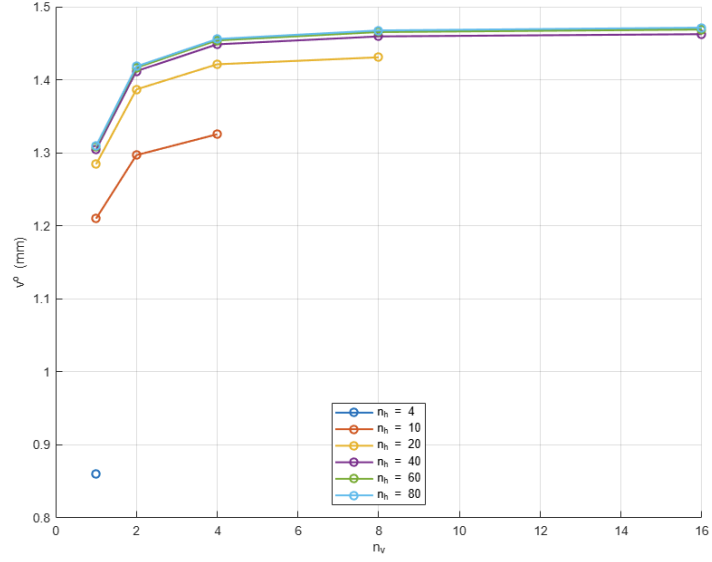


Figure 15: v^o vs n_v (Q4)

3.1.3 Ratio v^*/v^o

For the converged meshes, $v^* = 1.4974$ mm and $v^o = 1.4539$ mm. Thus, the ratio is:

$$v^*/v^o = 1.0299$$

3.1.4 Hoop stress

$\sigma_{\theta\theta}$ is the hoop stress around the edge of the hole. θ varies from $-\pi$ to π (as in Cartesian coordinate).

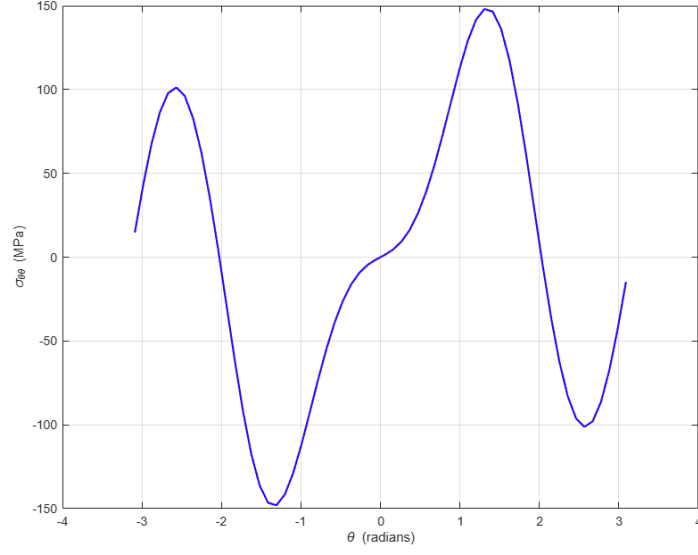


Figure 16: $\sigma_{\theta\theta}$ vs θ (Q4)

$$\sigma_{\theta\theta}^{max} = 148.0136 \text{ MPa at } \theta = 75 \text{ degrees}$$

θ is measured along the edge of the hole in the anti-clockwise direction from the horizontal line joining the center of the hole and the edge to its right. This location is at a distance of 65.1 mm from the fixed edge and 18.8 mm from the bottom edge of the cantilever beam.

3.1.5 Stress concentration factor

The stress concentration factor, $K = \sigma_{\theta\theta}^{max} / \sigma_{xx}^o$, where σ_{xx}^o is the xx component of the stress for a hole-free beam, taken at the point where $\sigma_{\theta\theta}^{max}$ occurs in the punctured beam.

For the hole-free beam (converged),

$$\sigma_{xx}^o = 54.7486 \text{ MPa}$$

Hence,

$$K = 2.7035$$

3.1.6 Original and deformed hole shapes

Deformation scaling factor = 40

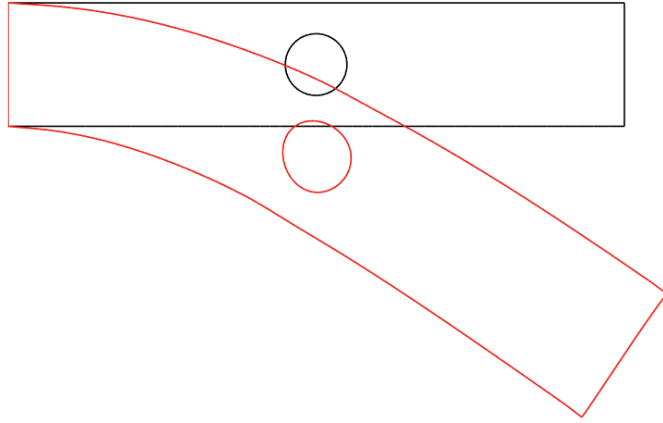


Figure 17: Original and deformed beam shape (Q4)

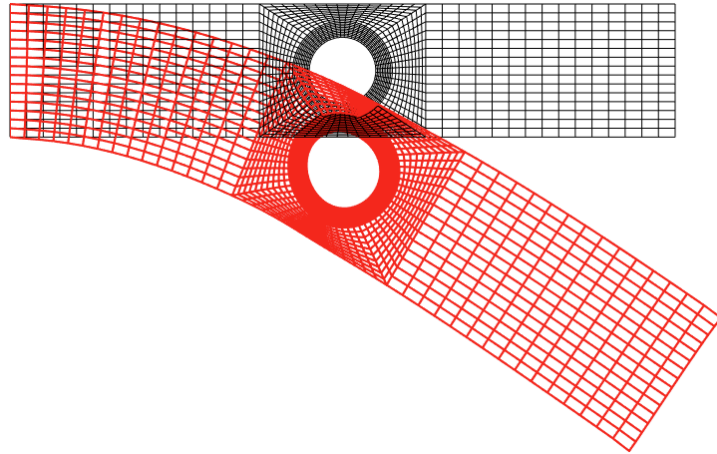


Figure 18: Original and deformed beam shape with mesh (Q4)

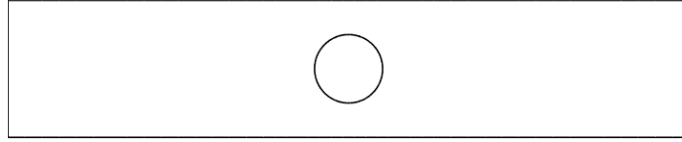


Figure 19: Original beam shape (Q4)

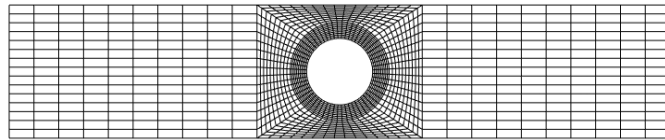


Figure 20: Original beam shape with mesh (Q4)

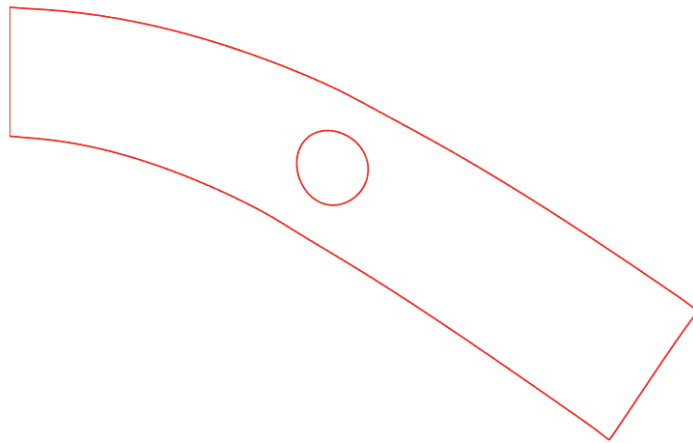


Figure 21: Deformed beam shape (Q4)

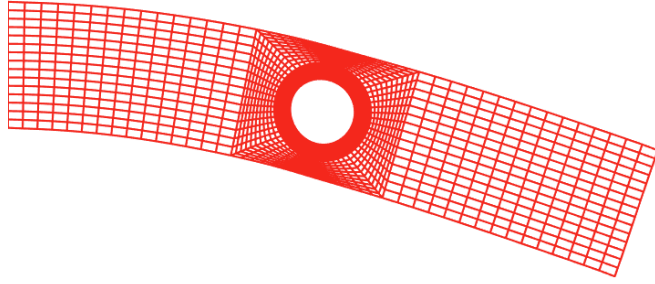


Figure 22: Deformed beam shape with mesh (Q4)

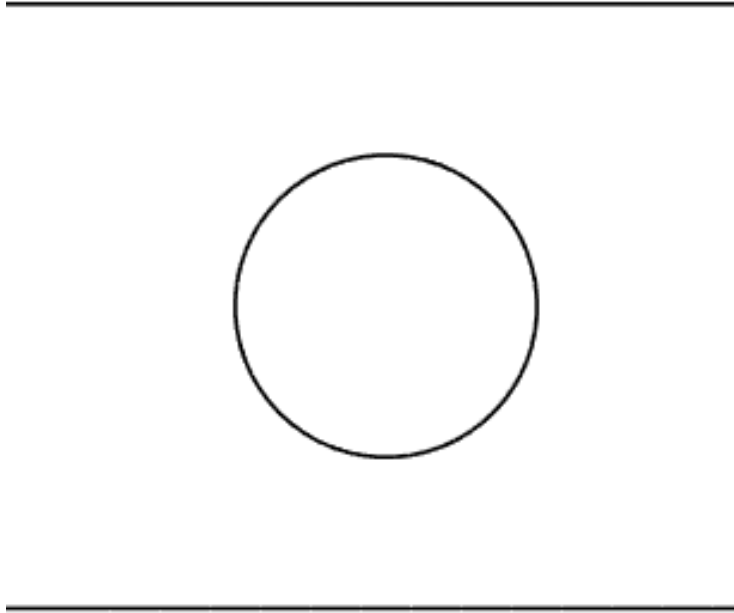


Figure 23: Original hole shape (Q4)

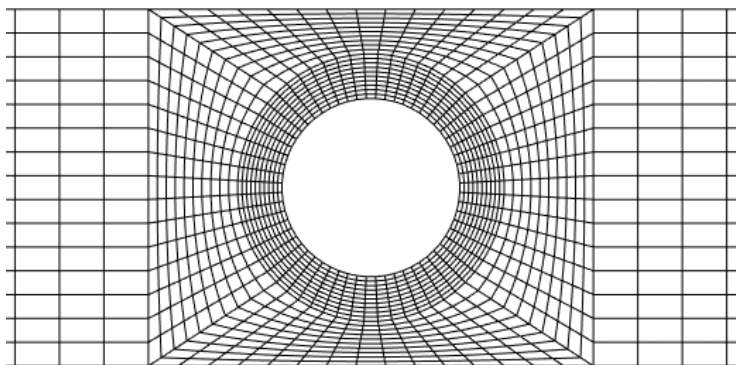


Figure 24: Original hole shape with mesh (Q4)

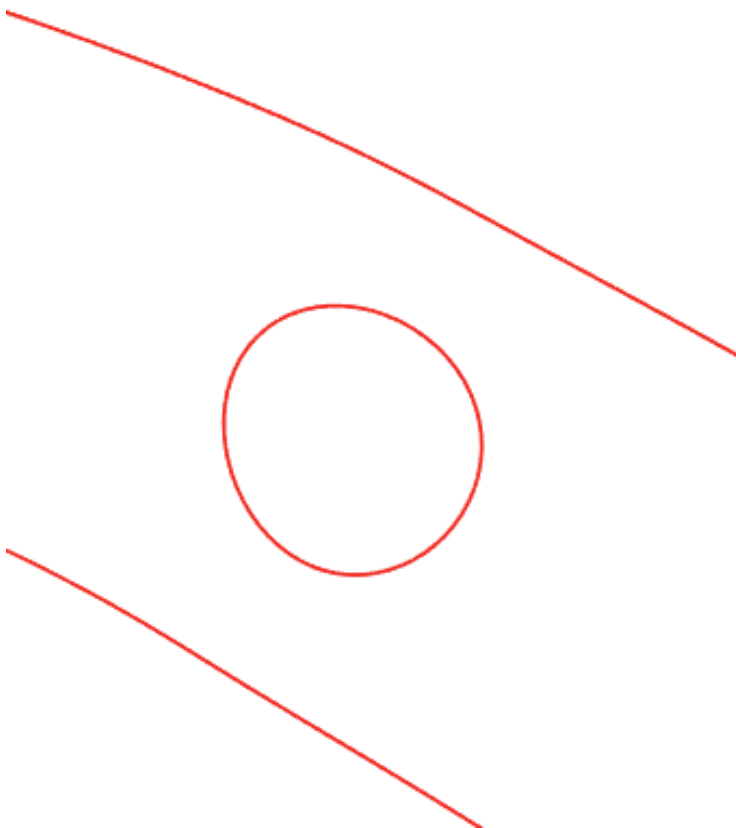


Figure 25: Deformed hole shape (Q4)

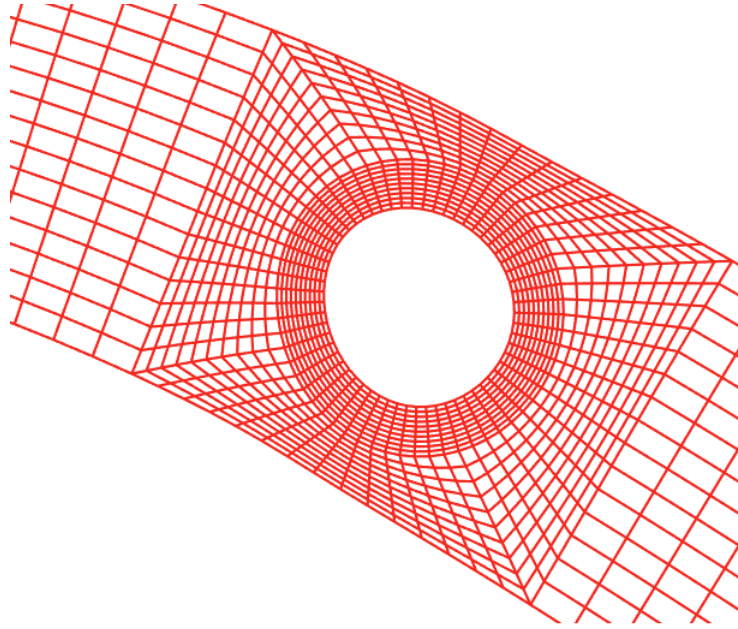


Figure 26: Deformed hole shape with mesh (Q4)

3.1.7 von Mises stress

von Mises stress, σ_{vm} is plotted. The values are in MPa.

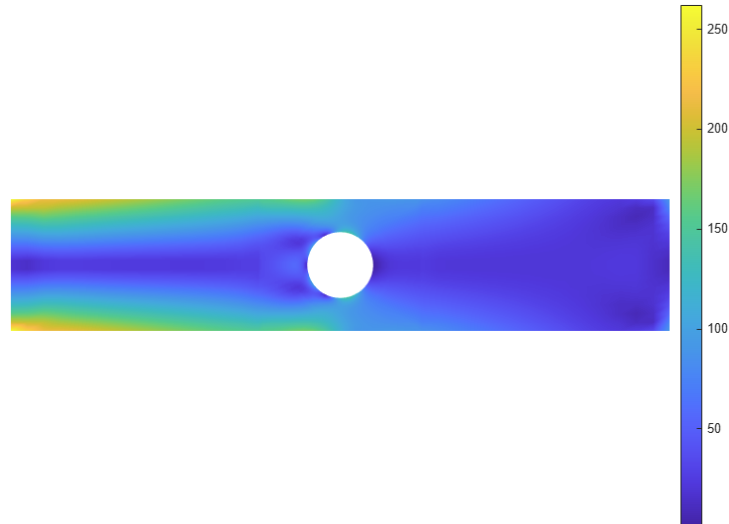


Figure 27: von Mises stress contour plot (Q4)

3.2 Isoparametric T6

3.2.1 Convergence study of beam with hole

N_h is the number of elements along the edge of the hole. N_{el} is the total number of elements. v^* is the tip selection of the bottom-right corner of the beam. $\sigma_{\theta\theta}^{max}$ is the maximum hoop stress along the edge of the hole. Highlighted row denotes converged mesh. v^* and $\sigma_{\theta\theta}^{max}$ are plotted against N_h and N_{el} to find converged mesh.

It is observed that the values of v^* are close to a significant number of decimal places. This implies that the starting mesh itself gives a close-to-converged result.

N_h	N_{el}	v^* (mm)	$\sigma_{\theta\theta}^{max}$ (MPa)
8	890	1.509881	131.4982
12	994	1.509965	141.2667
20	1084	1.509967	146.4921
40	1124	1.509976	150.5907
60	1240	1.509980	150.4616
80	1362	1.509978	150.6729

Table 4: Mesh convergence study for beam with hole (T6)

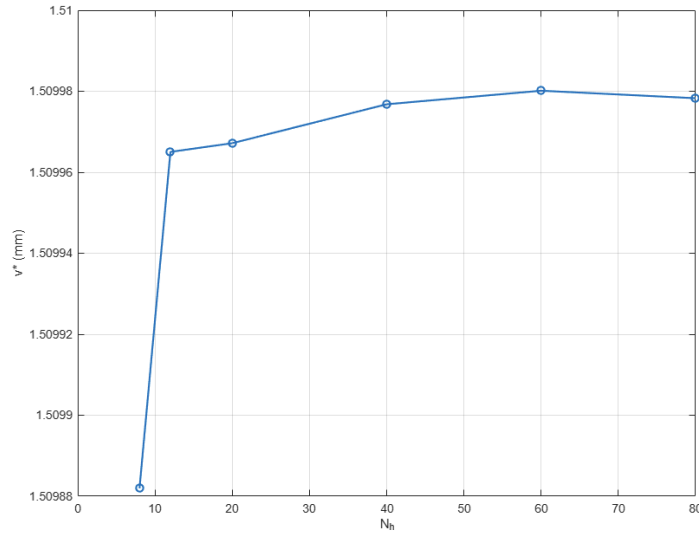


Figure 28: v^* vs N_h (T6)

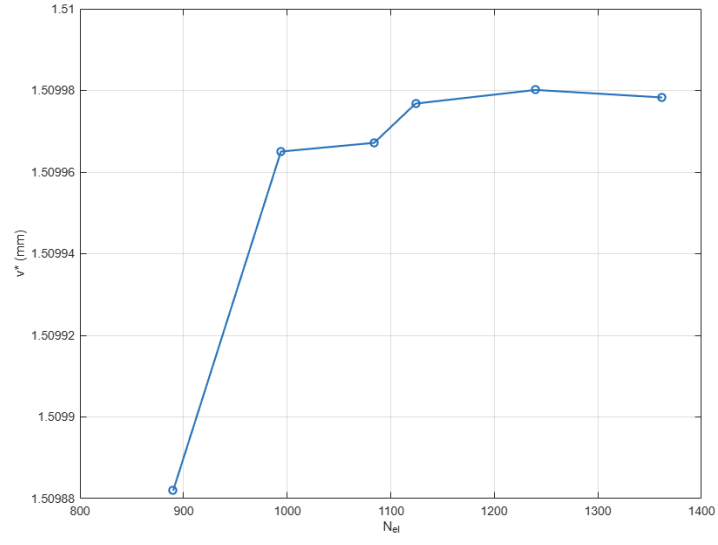


Figure 29: v^* vs N_{el} (T6)

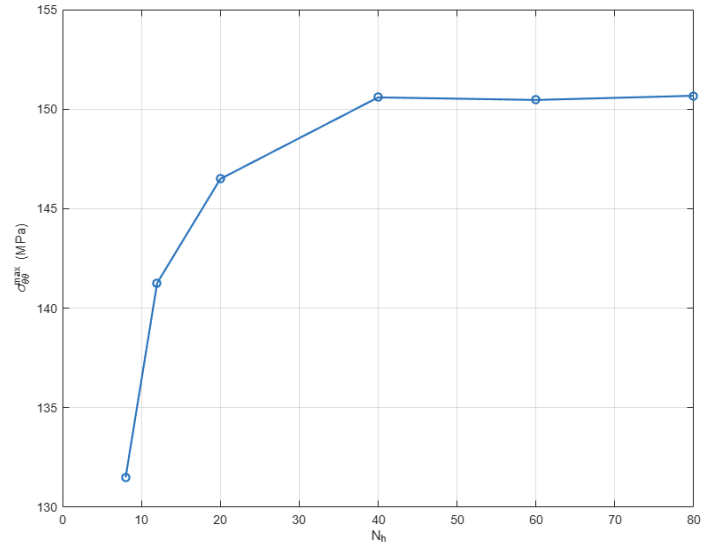


Figure 30: $\sigma_{\theta\theta}^{max}$ vs N_h (T6)

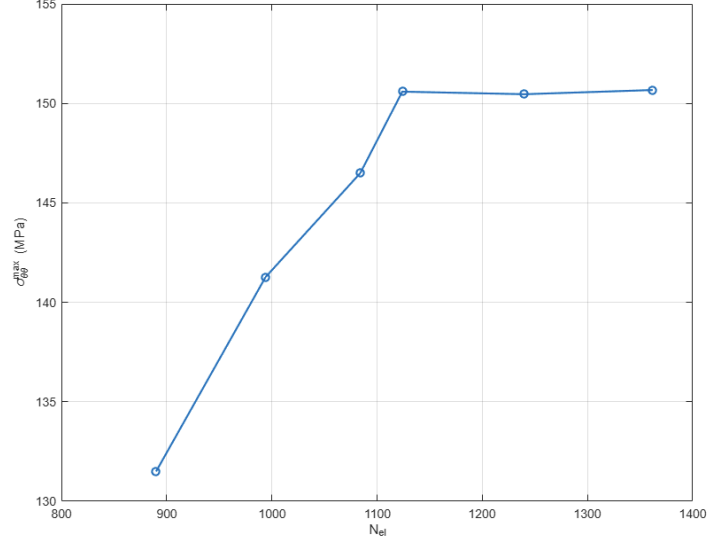


Figure 31: $\sigma_{\theta\theta}^{max}$ vs N_{el} (T6)

3.2.2 Convergence study of hole-free beam

n_h is the number of elements along the horizontal edge. n_v is the number of elements along the vertical edge. v^o is the deflection of the bottom-right corner of the beam. Highlighted row denotes the converged mesh. v^o is plotted against n_v for every n_h to find converged mesh.

n_h	n_v	v^o (mm)
4	1	1.455118
10	1	1.465069
	2	1.468328
	4	1.468715
20	1	1.468171
	2	1.469881
	4	1.472590
	8	1.472993
40	1	1.469400
	2	1.472030
	4	1.473781
	8	1.475481
	16	1.475914
60	1	1.469545
	2	1.472286
	4	1.474665
	8	1.476436
	16	1.477383

Table 5: Mesh convergence study for beam without hole (T6)

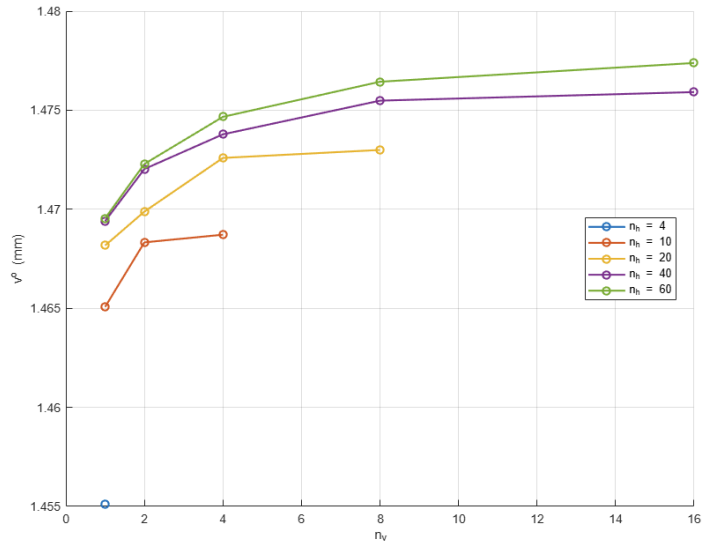


Figure 32: v^o vs n_v (T6)

3.2.3 Ratio v^*/v^o

For the converged meshes, $v^* = 1.509976$ mm and $v^o = 1.472286$ mm. Thus, the ratio is:

$$v^*/v^o = 1.0256$$

3.2.4 Hoop stress

$\sigma_{\theta\theta}$ is the hoop stress around the edge of the hole. θ varies from $-\pi$ to π (as in Cartesian coordinate).

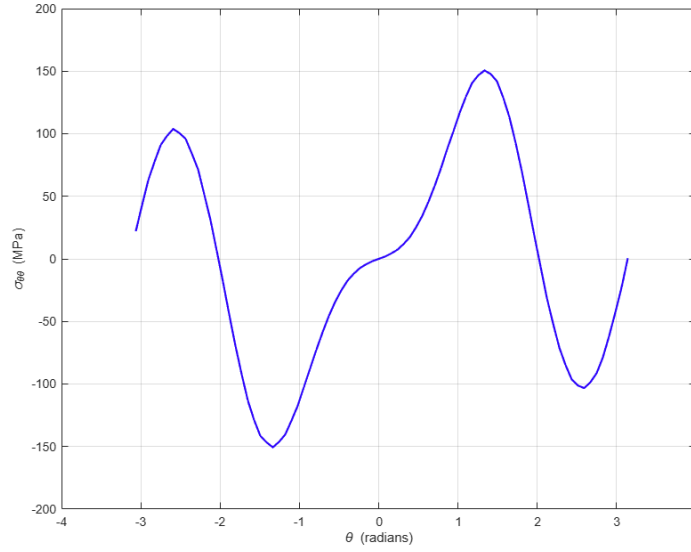


Figure 33: $\sigma_{\theta\theta}$ vs θ (T6)

$$\sigma_{\theta\theta}^{max} = 150.5907 \text{ MPa at } \theta = 76.50 \text{ degrees}$$

θ is measured along the edge of the hole in the anti-clockwise direction from the horizontal line joining the center of the hole and the edge to its right. This location is at a distance of 65 mm from the fixed edge and 18.8 mm from the bottom edge of the cantilever beam.

3.2.5 Stress concentration factor

The stress concentration factor, $K = \sigma_{\theta\theta}^{max} / \sigma_{xx}^o$, where σ_{xx}^o is the xx component of the stress for a hole-free beam, taken at the point where $\sigma_{\theta\theta}^{max}$ occurs in the punctured beam.

For the hole-free beam (converged),

$$\sigma_{xx}^o = 55.4198 \text{ MPa}$$

Hence,

$$K = 2.7172$$

3.2.6 Original and deformed hole shapes

Deformation scaling factor = 40

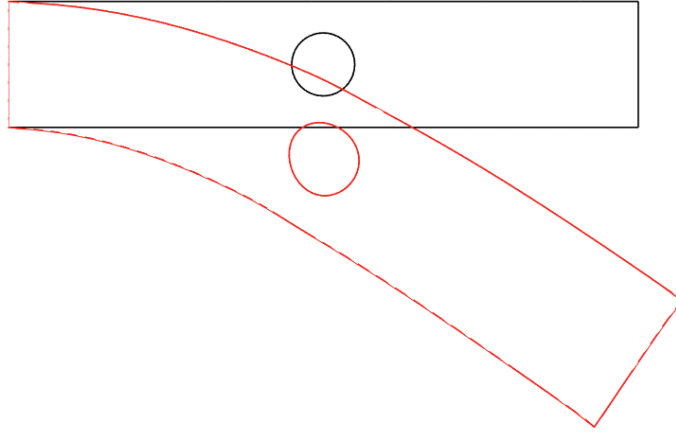


Figure 34: Original and deformed beam shape (T6)

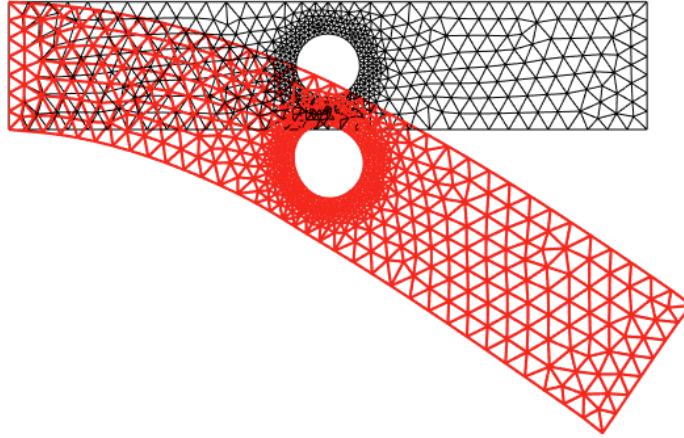


Figure 35: Original and deformed beam shape with mesh (T6)



Figure 36: Original beam shape (T6)

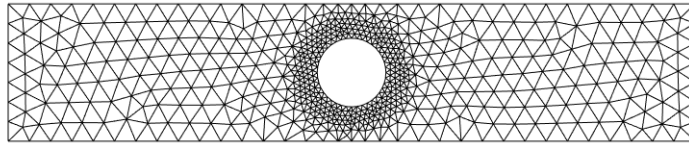


Figure 37: Original beam shape with mesh (T6)

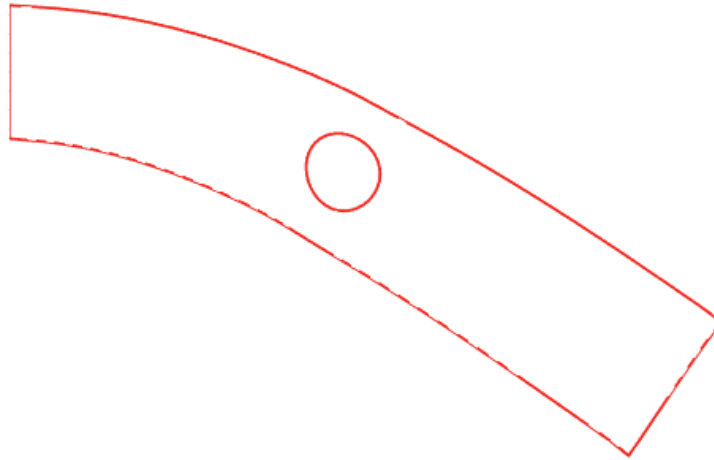


Figure 38: Deformed beam shape (T6)

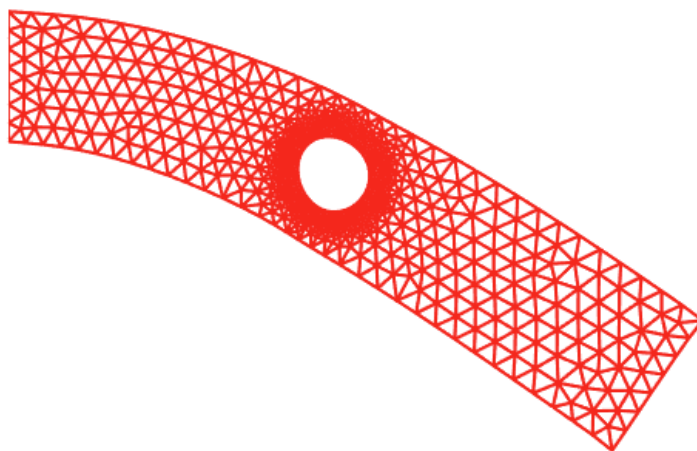


Figure 39: Deformed beam shape with mesh (T6)

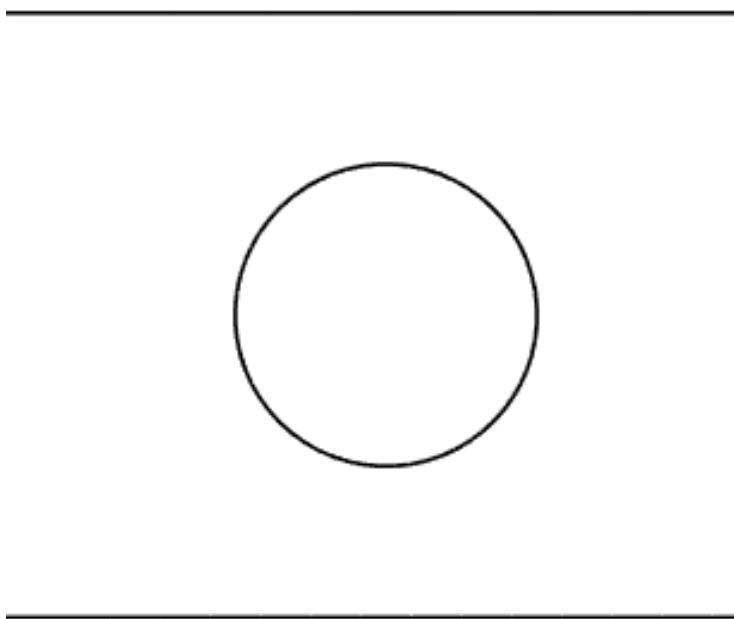


Figure 40: Original hole shape (T6)

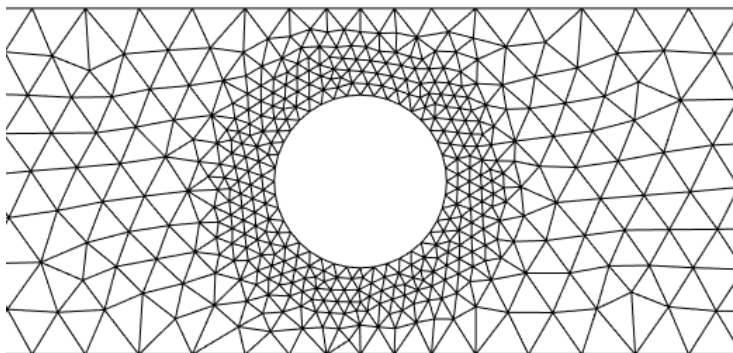


Figure 41: Original hole shape with mesh (T6)

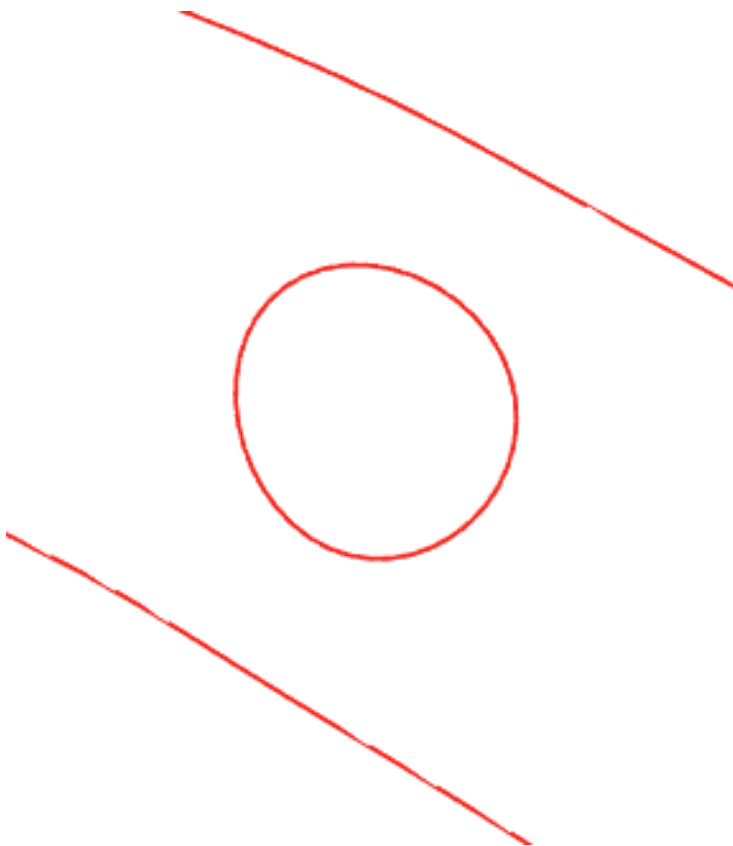


Figure 42: Deformed hole shape (T6)

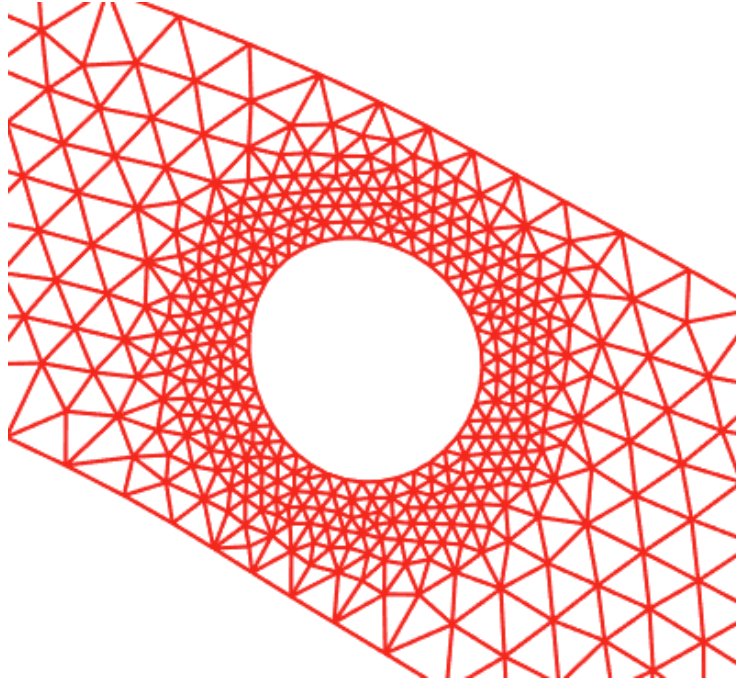


Figure 43: Deformed hole shape with mesh (T6)

3.2.7 von Mises stress

von Mises stress, σ_{vm} is plotted. The values are in MPa.

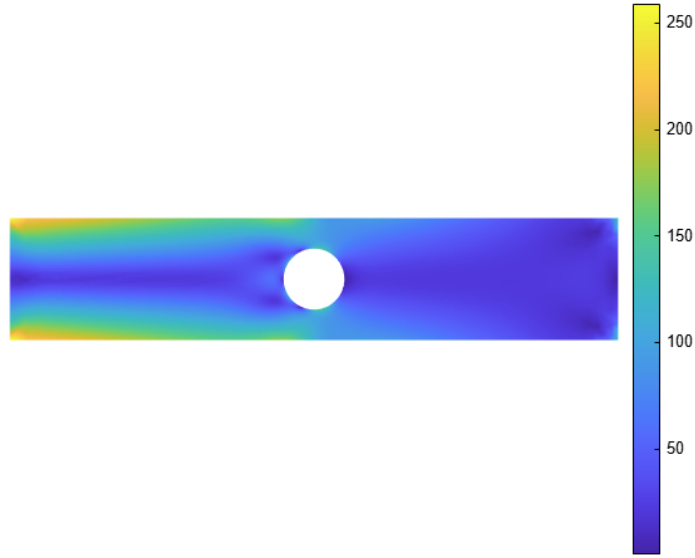


Figure 44: von Mises stress contour plot (T6)

3.3 Element suitability and convergence rate

Element Type	Min. elements for converged v^*	Min. elements converged $\sigma_{\theta\theta}^{max}$
Q4	1650	2200
T6	994	1124

Table 6: Summary Table

It is concluded from the summary table that tip deflection converges at a faster rate than stress convergence for both types of elements.

However, the tip and stress convergence rates are faster for T6 element than Q4. The convergence happens at a smaller number of elements. T6 element is also able to capture the stress concentration better than Q4. Hence, T6 element type is more suitable for this problem.

4 MATLAB Code

4.1 Isoparametric Q4

- *ConvergedHoleQ4.m* - produces convergence study plot for beam with hole.

- *ConvergedNoHoleQ4.m* - produces convergence study plot for beam without hole.
- *elemStressGPQ4.m* - computes stress at Gauss points of an element.
- *elemStressNodalQ4.m* - computes stress at nodes of an element
- *extractQ4.m* - extracts nodal coordinate matrix *nodM*, connectivity matrix *conM* and nodes on the edge of beam *lineElemM*.
- *findElementQ4.m* - finds the element in which (x, y) coordinate lies and its corresponding mapped coordinate (ξ, η) .
- *isoparamQ4stiffness.m* - compute stiffness matrix of an element.
- *jacobianCheckQ4.m* - checks Jacobian J for a typical Q4 element.
- *plotDeformedBeamQ4.m* - plots deformed beam w.o. mesh.
- *plotDeformedMeshQ4.m* - plots deformed mesh.
- *plotSavedMeshQ4.m* - plots original mesh.
- *SolverQ4.m*
- *visualizePointInMesh.m* - visualizes (x, y) coordinate in the beam.

4.2 Isoparametric T6

- *ConvergedHoleT6.m* - produces convergence study plot for beam with hole.
- *ConvergedNoHoleT6.m* - produces convergence study plot for beam without a hole.
- *ConvergedVerifyT6.m* - produces convergence study plot for HW4 problem
- *elemStressGPT6.m* - computes stress at Gauss points of an element.
- *elemStressNodalT6.m* - compute stress at nodes of an element
- *extractT6.m* - extracts nodal coordinate matrix *nodM*, connectivity matrix *conM* and nodes on the edge of beam *lineElemM*.
- *findElementT6.m* - finds the element in which (x, y) coordinate lies and its corresponding mapped coordinate (r, s) .
- *isoparamT6stiffness.m* - computes stiffness matrix of an element.
- *jacobianCheckT6.m* - checks Jacobian J for a typical T6 element.
- *plotDeformedBeamT6.m* - plots deformed beam w.o. mesh.
- *plotDeformedMeshT6.m* - plots deformed mesh.

- *plotSavedMeshT6.m* - plots original mesh.
- *SolverT6.m*
- *visualizePointInMesh.m* - visualizes (x, y) coordinate in the beam.